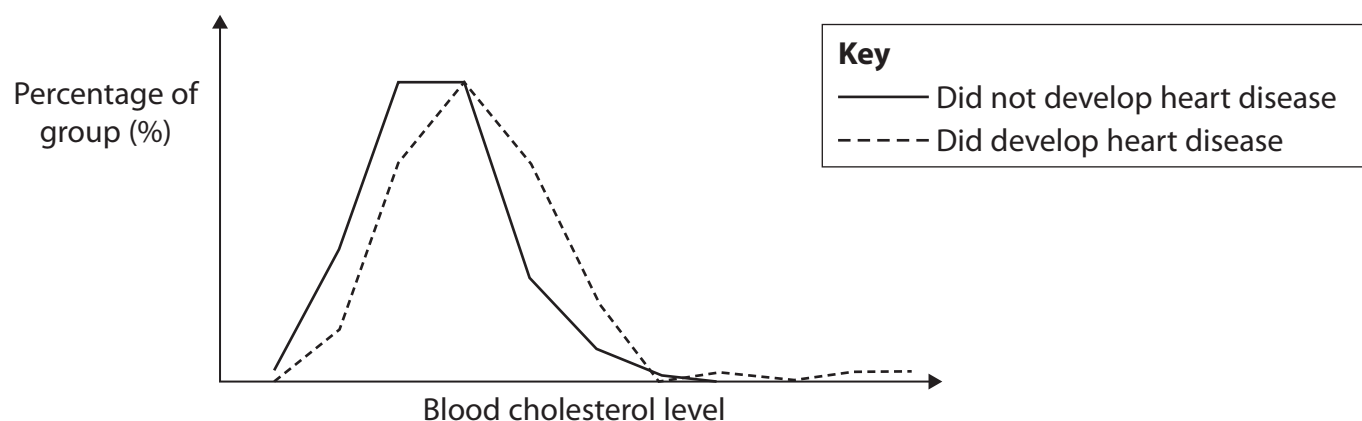


7 A number of factors increase the risk of cardiovascular disease (CVD), including blood cholesterol levels.

- (a) In one investigation, scientists measured the blood cholesterol level in a group of people and monitored the development of heart disease in the following 10 years.

The graph shows the results of this investigation.



Some people interpreted the data as showing evidence for a relationship between blood cholesterol levels and heart disease.

Other people interpreted the data as not showing evidence for a relationship between blood cholesterol levels and heart disease.

Describe the evidence for and against a relationship between blood cholesterol levels and heart disease.

(2)

Evidence for

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Evidence against

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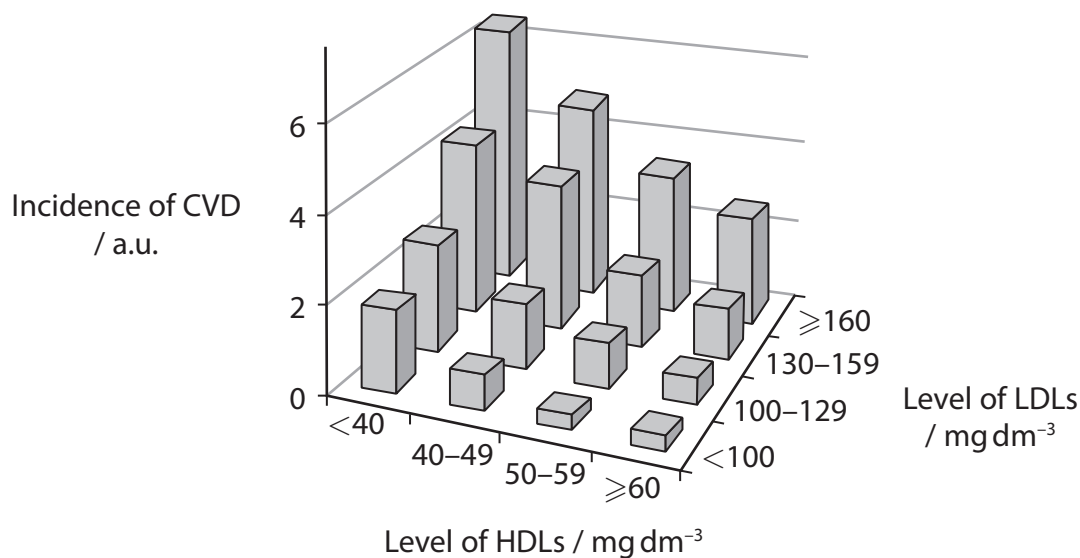
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- (b) Other investigations have shown that the levels of high-density lipoproteins (HDLs) and low-density lipoproteins (LDLs) are significant in determining the risk of CVD.

The graph shows the relationship between levels of HDLs and LDLs on the incidence of CVD.



The following conclusions can be made from this graph:

- as the levels of LDLs increases so does the incidence of CVD
- as the levels of HDLs increases the incidence of CVD decreases
- the higher the ratio of HDLs to LDLs the lower the incidence of CVD.

Describe the evidence shown in this graph that support these conclusions.

(3)

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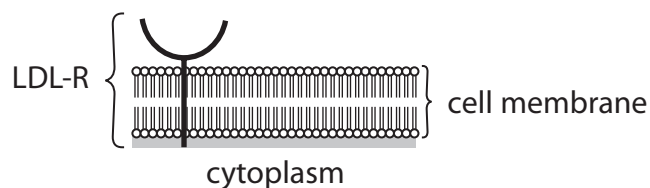
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- (c) The low-density lipoprotein receptor (LDL-R) is a protein on the surface of liver cells that is involved in the endocytosis of LDLs.

The diagram shows an LDL-R in a cell membrane.



- (i) Explain how the LDL-R is involved in endocytosis of LDLs.

(3)

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- (ii) The lower pH inside the cell causes the LDL to separate from its receptor.

Suggest how a change in pH could cause this separation.

(2)

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- (iii) Mutations in the gene coding for the LDL-R can result in an increased risk of CVD.

Explain why mutations in this gene can increase the risk of CVD.

(3)

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(Total for Question 7 = 13 marks)

